Exact Obstructions in Reconfiguring Nowhere-Zero Flows toward Five-Cycle Double Covers

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Status and AI-use disclosure. The standard five-cycle double cover conjecture remains open. This manuscript proves difference-rank and four-move normal forms, a short packing-to-switch lemma, and exact finite counterexamples only to proposed auxiliary flow-reconfiguration lemmas. It proves neither FiveCDC nor its negation.
OpenAI Codex agents using GPT-5-series models, under human direction, formulated the auxiliary questions and lemmas, found the finite examples, wrote the search and checking programs, performed the computations, drafted the proofs and this manuscript, and conducted an AI-assisted literature search. The "independent checker" is independent of the discovery implementation, not independent of AI. No human author, referee, or cited researcher is represented as having verified these results. Before submission, a human author must inspect every proof, independently rerun and preferably reimplement the enumerations, audit the graph and cover certificates, review the literature, and accept ordinary scholarly responsibility.

Abstract

Hušek and Šámal recently gave a component-parity condition on a nowhere-zero $\mathbb{F}_2^3$-flow which is equivalent, after choosing the flow, to the existence of a five-cycle double cover. Esperet et al. and Cranston et al. study reconfiguration of nowhere-zero flows by changes supported on cycles. We test the direct radius-one synthesis of these ideas.

For fixed-value moves on arbitrary Eulerian edge-sets, we first prove that unrestricted distance equals the rank of the edgewise difference values and give exact criteria for prescribed legal two- and four-move factorizations. A smallest $K_{3,3}$ example and a smallest cube bad-to-good example have unrestricted distance two but legal distance three. On a certified girth-ten non-Tait graph, a Jaeger-derived bad flow and a FiveCDC-derived good flow have difference rank three but exact legal Eulerian-support distance five. The five supports have 63 connected components in total, so for the connected-cycle adjacency in the cited reconfiguration papers the certificate proves only $5 \leq d \leq 63$.

We next prove a human-checkable bridge. If one value-class matching of a cubic $\mathbb{F}_2^3$-flow admits two edge-disjoint joins in its complement, then an explicitly determined even support, and therefore a finite sequence of legal simple-cycle switches, reaches a flow satisfying the Hušek–Šámal condition. Connectedness of that support is exactly what makes one switch sufficient.

We also analyze the residual blockers of one such join. Every blocker whose nonmatching cut has size three, five, or seven has an occurring partner-free flow direction, and cycle–cut orthogonality then gives a legal switch making that old shore terminal-even. Under the natural all-in condition we derive an exact aggregate-gain formula: from zero tight count, fixed-residual lexicographic descent occurs exactly when old even four-cuts are parity-protected and one global integer gain is positive. A certified girth-ten cyclically-4 Tait control shows that the reduced

cut-space coordination obstruction can still block all three directions of one $(5, 1)$ shore. Thus the local theorem is unconditional but the non-Tait global coordination step remains open.

We then give an exact 26-vertex obstruction to radius one. The graph is simple, cubic, cyclically 4-edge-connected, has girth five, and is not Tait-colourable. One displayed value class already admits two edge-disjoint boundary joins. Nevertheless, for the displayed nowhere-zero flow, a standard-library checker exhausts all 9,213 simple cycles and all 1,485 legal cycle–value switches; none reaches the component condition. The packing construction splits into two circuit components, and switching both reaches the condition. Thus the displayed flow has exact distance two. An independently checked five-cycle double cover of the same graph confirms that this is not a counterexample to FiveCDC.

For calibration we record a cyclically 4-edge-connected, girth-five, Tait-colourable 40-vertex radius-one obstruction, and a 60-vertex positive control repaired by one 25-edge cycle. The latter is also a literal instance of the packing-to-switch lemma. All finite claims are accompanied by solver-independent, certificate-based checking code. Possible novelty is stated conservatively and remains subject to specialist prior-art review.

As a distinct calibration, we give a cyclically 4-edge-connected 40-vertex Jaeger-star state with exact augmented reciprocal-exchange distance three. This disproves a project-created radius-two shortcut, but uses a different state graph from simple-cycle switching. The same graph has an explicit FiveCDC.

1 Scope and relation to prior work

A *five-cycle double cover* (FiveCDC) of a finite graph is an indexed list $C_0, \ldots, C_4$ of Eulerian edge-subsets such that every edge lies in exactly two members. Eulerian means even degree at every vertex; neither connectedness nor 2-regularity is required. Empty members are allowed, so this is the usual "at most five" convention. Only the standard, unoriented conjecture is considered here.

Hušek and Šámal state FiveCDC as Conjecture 1.9 and give an equivalent flow-selection formulation as Conjecture 3.19 in [3]. In particular, their Theorem 3.16 supplies the component-parity criterion used below. Thus FiveCDC remains open, and the remaining universal issue is to select a suitable nowhere-zero $\mathbb{F}_2^3$-flow.

Cranston, Li, Su, Wang, and Xu define adjacency of nowhere-zero flows by requiring their difference to be supported on a cycle [2, 1]. In the exponent-two group $\mathbb{F}_2^3$, a difference supported on a simple cycle has one constant nonzero value around that cycle. The switches below are precisely this special case, with legality made explicit so that no zero edge is created. Cranston et al. also use an auxiliary relation $\mathcal{F}^*$ in which the difference may be supported on an even subgraph. Its different circuit components may carry different values. Our Eulerian-support move permits a disconnected support but requires one fixed value on all of it. Thus, for the relations used here,

$$\text{connected-cycle adjacency} \;\subseteq\; \text{fixed-value Eulerian-support adjacency} \;\subseteq\; \mathcal{F}^* \text{ adjacency.}$$

The exact distance-five lower bound below is only for the middle relation. It implies a lower bound for connected-cycle distance, but no lower bound for the broader $\mathcal{F}^*$ distance.

Our contribution has deliberately narrow scope:

1. an exact normal form for two fixed-value Eulerian-set moves, including smallest $K_{3,3}$ and cube obstructions to legal distance two;
2. an exact rank-three criterion for a prescribed four-move order, and a certified pair of flows with zero-allowing distance three but legal Eulerian-support distance five;
3. a direct proof that a two-join packing yields a finite legal switch sequence to a Hušek–Šámal-good flow;

4. the partner-free single-shore switch theorem and the exact aggregate-gain characterization for fixed-residual descent, together with a certified high-girth Tait control for the remaining reduced cut-space obstruction;

5. an exact strict 26-vertex countermodel to the assertion that one simple-cycle switch always suffices even when a value class already packs, together with an exact distance-two path and a positive FiveCDC certificate; and

6. one 40-vertex negative and one 60-vertex positive control; and

7. an exact 40-vertex countermodel to a separate radius-two reciprocal-exchange assertion for Jaeger-star tree states.

No reduction from FiveCDC to a universal radius-one assertion is claimed here, and no statement about orientable cycle double covers is made.

2 Flows, component defects, and switches

Put $W = \mathbb{F}_2^3$, written additively, and identify W with $\{0, \ldots, 7\}$ by binary coordinates. Let G be a finite simple cubic graph. A map

$$f : E(G) \longrightarrow W - \{0\}$$

is a nowhere-zero flow if

$$\sum_{e \ni v} f(e) = 0 \qquad (v \in V(G)).$$

At a cubic vertex the three incident values are nonzero and pairwise distinct. Consequently each value class

$$M_a(f) = \{e : f(e) = a\}, \qquad a \neq 0,$$

is a matching.

For a nonzero linear functional $\mu \in W^*$, identified with its binary dot-product vector, define

$$\begin{aligned} F_\mu(f) &= \{e : \mu(f(e)) = 1\}, \\ K_\mu(f) &= \{e : \mu(f(e)) = 0\} = E(G) - F_\mu(f). \end{aligned} \tag{2.1}$$

For $r \in W$ with $\mu(r) = 1$, let

$$Z_r(f) = \partial M_r(f)$$

be the endpoints of the matching $M_r(f)$. Define

$$d_\mu(f) = \#\left\{Q \in \mathrm{Comp}(V(G), K_\mu(f)) : |Q \cap Z_r(f)| \text{ is odd}\right\}. \tag{2.2}$$

Lemma 2.1 (affine-value independence)**.** *The component parities in (2.2) do not depend on the choice of r satisfying $\mu(r) = 1$.*

Proof. Fix a component Q of $K_\mu(f)$. The parity of $|Q \cap Z_r(f)|$ equals the parity of the number of r-valued edges in $\delta(Q)$, since an internal matching edge contributes two endpoints. Every edge in $\delta(Q)$ has μ-value one.

Write $\ker \mu = \{0, p, q, p+q\}$, so the four affine values are $r, r+p, r+q, r+p+q$. Let their boundary multiplicities modulo two be n_0, n_p, n_q, n_{p+q}. Summing the flow equations on Q gives

$$n_0 r + n_p(r+p) + n_q(r+q) + n_{p+q}(r+p+q) = 0.$$

Comparing coefficients in the basis (r, p, q) gives

$$n_0 + n_p + n_q + n_{p+q} = 0, \quad n_p + n_{p+q} = 0, \quad n_q + n_{p+q} = 0.$$

Hence all four multiplicities are equal. This proves the claim. $\square$

Definition 2.2. The flow f is *H–S-good* if $d_\mu(f) = 0$ for some $\mu \neq 0$. Its defect profile is

$$d(f) = (d_1(f), \ldots, d_7(f)).$$

By Theorem 3.16 and Observation 3.15 of [3], an H–S-good flow produces a standard FiveCDC. Their Conjecture 3.19 asserts that every bridgeless cubic graph has some H–S-good flow and is equivalent to FiveCDC.

Definition 2.3 (legal simple-cycle switch)**.** Let $a \in W - \{0\}$, and let C be a simple cycle of G such that $C \cap M_a(f) = \varnothing$. Define

$$f^{a,C}(e) = \begin{cases} f(e) + a, & e \in E(C), \\ f(e), & e \notin E(C). \end{cases} \tag{2.3}$$

The cycle equation preserves flow conservation, and $C \cap M_a(f) = \varnothing$ is exactly the condition preventing a zero value. We write $\mathrm{dist}_{\mathrm{HS}}(f)$ for the minimum number of legal simple-cycle switches needed to reach an H–S-good flow, when such a sequence exists.

3 A rank-two normal form for fixed-value Eulerian moves

For this section only, permit one move to have an arbitrary Eulerian edge-set C, possibly disconnected: choose $a \in W - \{0\}$ and replace f by $f + a1_C$. Call the move legal when the new flow is nowhere-zero. This is a coarser one-step relation than the simple-cycle relation above. Every legal fixed-value Eulerian move nevertheless decomposes into legal simple-cycle switches by splitting C into edge-disjoint circuits: during the decomposition, every edge has either its old nonzero value or its final nonzero value.

Theorem 3.1 (difference-rank normal form)**.** *Let f, g be nowhere-zero W-flows, put $h = f + g$, and let*

$$D = \mathrm{span}\{h(e) : e \in E(G)\}.$$

If zero-valued intermediate flows are permitted, the minimum number of fixed-value Eulerian moves from f to g is $\dim D$.

Proof. If k moves use constants $a_1, \ldots, a_k$, every value $h(e)$ lies in their span, so $k \geq \dim D$. Conversely, choose a basis $a_1, \ldots, a_r$ of D and write

$$h(e) = \sum_{i=1}^{r} \lambda_i(e) a_i.$$

Since h is a flow, each binary coordinate $\lambda_i \circ h$ is a flow. Hence $C_i = \{e : \lambda_i(e) = 1\}$ is Eulerian. Adding a_i on C_i for $i = 1, \ldots, r$ adds exactly h. $\square$

The legality issue has a complete answer in rank two. Write

$$D - \{0\} = \{a, b, a + b\}, \qquad H_w = \{e : h(e) = w\}.$$

Theorem 3.2 (rank-two legal-order criterion). *For the ordered basis (a, b), the only two-move factorization is*

$$f \xrightarrow{a,\ H_a \cup H_{a+b}} f + a1_{H_a \cup H_{a+b}} \xrightarrow{b,\ H_b \cup H_{a+b}} g.$$

It is legal if and only if

$$H_{a+b} \cap f^{-1}(a) = \varnothing. \tag{3.1}$$

Consequently all six ordered bases are blocked if and only if all six directed transitions

$$(f(e), g(e)) = (u, v), \qquad u, v \in D - \{0\}, \quad u \neq v,$$

occur.

Proof. The two supports are the unique binary coordinate supports of h in the basis (a, b), and hence are Eulerian. An edge of H_a cannot have $f(e) = a$, since then $g(e) = 0$. Thus the first intermediate flow vanishes exactly on the set in (3.1). The second endpoint is the nowhere-zero flow g. Finally, a blocker in (3.1) has $(f(e), g(e)) = (a, b)$; varying the ordered basis gives precisely the six displayed transitions. □

These statements are elementary but useful in delimiting the reconfiguration strategy. On $K_{3,3}$, with graph6 record `EFz_`, the flows

$$f = (1, 2, 3, 2, 3, 1, 3, 1, 2),$$
$$g = (1, 3, 2, 2, 1, 3, 3, 2, 1)$$

in the canonical edge order have difference $(0, 1, 1, 0, 2, 2, 0, 3, 3)$ and realize all six directed transitions. Thus their unrestricted distance is two but their legal distance is three. A three-step path is obtained by adding constants $4, 3, 6$ on the circuits

$$\{1, 2, 4, 5\}, \quad \{1, 2, 7, 8\}, \quad \{1, 2, 4, 5\},$$

respectively. This is smallest among connected simple cubic graphs; the only order-four graph is K_4, where the six-transition pattern is impossible. The supplied checker also exhausts all 210 flows of K_4.

A stronger bad-to-good example occurs on the cube `G?zTb_`. It has a displayed H–S-bad flow f and H–S-good flow g with rank-two difference, the same nonempty binary coordinate circuit, all six two-move orders blocked, and a legal three-circuit path. Exact enumeration gives 5,712 nowhere-zero flows on the cube, of which 1,344 are H–S-bad and 4,368 are H–S-good. All flows on the three connected simple cubic graphs through order six are H–S-good, so the cube example is the smallest bad-to-good instance in that domain. Both graphs are Tait-colourable and have explicit FiveCDCs; they obstruct only the two-move shortcut.

The complete edge tables, component defects, three-step paths, small-order proof, machine-readable witnesses, and standard-library checker are in `scratch/flow-rank-two-legal-order-20260729/`.

4 A rank-three four-move normal form

The rank-two criterion is edge-local because the move memberships are unique. With four ordered constants spanning W, every edge has two possible membership words. Their common choice variable is governed by one binary incidence system.

Theorem 4.1 (rank-three four-move criterion)**.** *Let G be a finite multigraph, with loops counted twice in incidence sums, and let f, g be nowhere-zero W-flows. Fix an ordered four-tuple*

$$(a_1, a_2, a_3, a_4) \in (W - \{0\})^4$$

spanning W, and define

$$A : \mathbb{F}_2^4 \longrightarrow W, \qquad Az = \sum_{j=1}^{4} z_j a_j.$$

Let d be the unique nonzero member of $\ker A$, put $h = f + g$, and choose one preimage $z^0(e) \in A^{-1}(h(e))$ for every edge. At each vertex write

$$\sum_{e \ni v} z^0(e) = q_v d, \qquad q_v \in \mathbb{F}_2. \tag{4.1}$$

For an edge e, let $L_e \subseteq \mathbb{F}_2$ contain precisely the choices y for which all four prefixes

$$f(e) + \sum_{j=1}^{k} \big(z_j^0(e) + y d_j\big) a_j, \qquad 1 \le k \le 4. \tag{4.2}$$

are nonzero.

There is a legal realization using the four constants in the prescribed order if and only if bits $y_e \in L_e$ solve

$$By = q. \tag{4.3}$$

where B is the unoriented vertex-edge incidence matrix over $\mathbb{F}_2$.

Equivalently, reject if some L_e is empty. Otherwise force $y_e = 1$ on $P = \{e : L_e = \{1\}\}$, leave $F = \{e : L_e = \{0, 1\}\}$ free, and put $q' = q + B1_P$. Equation (4.3) is soluble exactly when every connected component of the spanning subgraph (V, F), including isolated vertices, contains an even number of vertices on which q' is one.

Proof. Flow conservation gives

$$A \sum_{e \ni v} z^0(e) = \sum_{e \ni v} h(e) = 0,$$

so the sum lies in $\ker A = \langle d \rangle$, proving that (4.1) is defined. Every preimage of $h(e)$ is uniquely $z^0(e) + y_e d$. Thus the four supports are Eulerian precisely when

$$0 = \sum_{e \ni v} \big(z^0(e) + y_e d\big) = \big(q_v + (By)_v\big) d$$

at every vertex. Since $d \neq 0$, this is (4.3); condition (4.2) is exactly edgewise legality.

After substituting the forced-one variables, the remaining system is $B_F y_F = q'$. Incidence columns generate exactly the vectors of even parity on each free-edge component. Necessity is the handshake lemma. For sufficiency, choose a spanning tree in each component and eliminate the demands from its leaves toward a root. Loop columns are zero, and parallel edges remain separate columns, so the same proof covers both. □

Proposition 4.2 (an exact distance-five pair)**.** *There is a simple connected cubic non-Tait graph G of order 890 and girth ten, with nowhere-zero W-flows f, g, such that:*

1. *f is H–S-bad and g is induced by an explicit FiveCDC and is H–S-good;*

2. *the values of* $f+g$ *span* W*, so the zero-allowing fixed-value Eulerian-support distance is three;*

3. *the legal fixed-value Eulerian-support distance from* f *to* g *is exactly five; and*

4. *for connected simple-cycle adjacency,* $5 \le \operatorname{dist}(f,g) \le 63$.

Proof. Take the Foster graph with LCF string

$$[17,-9,37,-37,9,-17]^{15}.$$

Delete vertex 0, and substitute the resulting 89-vertex three-pole for every vertex of the Petersen graph. The checker reconstructs the resulting 890-vertex, 1335-edge graph and verifies simplicity, cubicity, girth ten, edge-connectivity three, cyclic edge-connectivity three, and non-Tait-colourability. The last property also has a short reduction: parity in each odd-order pole forces its three boundary colours to be distinct, which would contract to a Tait colouring of the Petersen graph.

The frozen Jaeger state gives a nowhere-zero flow f with odd-kernel sizes

$$(506, 507, 526)$$

and H–S profile

$$(174, 192, 138, 184, 130, 140, 110).$$

Its seven entries are positive. The displayed FiveCDC, after assigning its five coordinates the points $(0,1,2,3,4)$ of W, gives a flow g with profile

$$(142, 160, 132, 0, 148, 130, 150).$$

The difference values have rank three, and Theorem 3.1 gives zero-allowing distance three. All 168 ordered bases are illegal. The independent implementation of Theorem 4.1 exhausts all 1848 ordered spanning four-tuples: 1176 have an empty local list L_e, and the remaining 672 fail the free-component parity condition. Thus no legal path has at most four moves.

Five legal moves use constants

$$(5, 1, 6, 2, 7)$$

on Eulerian supports of sizes

$$(566, 608, 565, 610, 604).$$

The bitmasks in the certificate verify every vertex parity and every intermediate nonzero value and end exactly at g. This proves Eulerian-support distance five. Those supports have respectively

$$(14, 11, 10, 14, 14)$$

connected components. Each is a simple cycle, and the components of a legal Eulerian support may be switched sequentially in any order. Hence the same data give a 63-step connected-cycle path. Every connected-cycle move is an Eulerian-support move, which gives the lower bound five. □

The source and independently written standard-library checkers, exact flows, five support masks, component lists, and full proof ledger are in

`scratch/petersen-foster-girth10-nontait-20260729/`

and

`scratch/rank3-fourmove-normal-form-20260729/`.

This proposition is a distance statement for one selected pair. It does not say that f is five moves from the set of all H–S-good flows, does not give a distance-five lower bound in $\mathcal{F}^*$, and does not advance the existential flow-selection conjecture.

5 A human packing-to-switch lemma

For an edge set J, write ∂J for the set of vertices having odd degree in J. A T-join is an edge set J with $\partial J = T$.

Theorem 5.1 (packing-to-switch)**.** *Fix $a \neq 0$ and $\mu(a) = 1$. Put*

$$F = F_\mu(f), \qquad M = M_a(f), \qquad J_0 = F - M.$$

Suppose $G - M$ contains two edge-disjoint ∂M-joins J_1, J_2. Then switching a on

$$C = J_0 \triangle J_1 \tag{5.1}$$

produces an H–S-good flow. The support C is a disjoint union of legal simple cycles, so this operation is a legal finite sequence of simple-cycle switches. If C is connected, one switch suffices.

Proof. The binary characteristic function $\mu \circ f$ is a flow, so F is Eulerian. Because $M \subseteq F$,

$$\partial J_0 = \partial(F - M) = \partial M.$$

Thus J_0 and J_1 have the same boundary, and $C = J_0 \triangle J_1$ is Eulerian. Both avoid M, hence so does C.

In a cubic graph every vertex has degree zero or two in an Eulerian edge set. Every nonempty connected component of C is therefore a simple cycle. Switch a on all these components. Since $\mu(a) = 1$, the new affine support is

$$\begin{aligned} F \triangle C &= F \triangle \big((F - M) \triangle J_1\big) \\ &= M \cup J_1. \end{aligned} \tag{5.2}$$

The union is disjoint because $J_1 \subseteq E(G - M)$.

The matching M_a itself is unchanged: the support avoids M_a, and an edge with old value different from a cannot acquire value a after adding a, since that would force its old value to be zero. Consequently every component switch remains legal when the components are performed sequentially.

Finally, J_2 avoids both M and J_1, so by (5.2) it lies in the new zero-side K_μ. Restricting the equality $\partial J_2 = \partial M$ to any component Q of this zero-side and applying the handshaking lemma gives

$$|Q \cap \partial M| \equiv 0 \pmod 2.$$

By Lemma 2.1, this is the H–S condition for μ. $\square$

Remark 5.2. The theorem exposes the radius-one gap. A packing gives an Eulerian switch support canonically, but that support may have several cycle components. The theorem guarantees a finite reconfiguration sequence; it guarantees one step only when the support is connected.

6 Residual blockers and an exact aggregate-gain gate

Fix a nonzero target value b, put $M = M_b(f)$, $H = G - M$, and let $J \subseteq H$ be a $T = \partial M$-join. Then

$$\mathcal{F} = M \,\dot{\cup}\, J$$

is Eulerian. For a component Q of $G-\mathcal{F}$, put

$$k_Q = |\delta_H(Q)| = |\delta_J(Q)|, \qquad q_Q = |\delta_M(Q)|, \qquad d_Q = k_Q + q_Q.$$

Call Q odd when q_Q is odd. Summing the flow equations over its shore gives

$$\bigoplus_{e\in\delta_H(Q)} f(e) = b. \tag{6.1}$$

In particular k_Q is odd and at least three. The profile $(k_Q, q_Q) = (3,1)$ is called tight. The coordinatewise-minimal non-tight odd profiles are $(3,3)$ and $(5,1)$.

For fixed b, the six values outside $\{0,b\}$ form three partner pairs $\{u, u+b\}$. An occurring value t on $\delta_H(Q)$ is *partner-free* when $b+t$ does not occur there.

Lemma 6.1 (partner-free direction)**.** *If $k_Q \in \{3,5,7\}$, then $\delta_H(Q)$ has a partner-free occurring value. If $k_Q = 3$, every occurring value is partner-free. The bound is sharp for this criterion at $k_Q = 9$.*

Proof. Quotient W by $\langle b\rangle$. The three partner pairs map to the three nonzero vectors $c_1, c_2, c_3 \in \mathbb{F}_2^2$, with $c_1+c_2+c_3 = 0$. If n_i is the total multiplicity in the i-th pair, the projection of (6.1) says that the three parities n_i mod 2 are equal. Their sum is odd, so all three are odd. If no occurring value were partner-free, both members of every pair would occur; then every n_i would be an odd integer at least three, forcing $k_Q \geq 9$. When $k_Q = 3$, every $n_i = 1$. Sharpness is witnessed, for $b = 1$, by the multiset $2,2,3,4,4,5,6,6,7$. ☐

Theorem 6.2 (single-shore parity kill)**.** *Let Q be odd and let t be partner-free on its H-cut. There is a legal binary t-switch support $X \subseteq G - M_t$ such that the old shore Q has even target-cut parity after the switch.*

Proof. Put $P_t = M_b \cup M_{b+t}$, $L = G - M_t$, and $S = \delta_M(Q)$. Partner-freeness gives $P_t \cap \delta_G(Q) = S$. On $Z_1(L;\mathbb{F}_2)$, consider $X \mapsto |X\cap S|$ mod 2.

If this functional vanished on the cycle space, cycle–cut orthogonality would give a shore U with $\delta_L(U) = S$. The full cut $\delta_G(U)$ would consist of the odd number q_Q of b-edges in S, together with some t-edges. Its flow xor would be b or $b+t$, both nonzero, contradicting the flow cut equation. Thus some binary cycle $X \subseteq L$ meets S oddly. It is a legal t-switch because it contains no t-valued edge. Under this switch, membership in the target class changes exactly on $P_t \cap X$, so the target-cut parity of Q changes by $|X\cap S| = 1$. ☐

The theorem concerns the old shore only. It is independent of how a new join is chosen, but it does not prevent a crossing or proper subshore from becoming odd.

We now record the exact fixed-residual calculation. Let $X \subseteq G - M_t$ be a legal switch and put

$$D^- = X \cap M_b, \qquad D^+ = X \cap M_{b+t}.$$

Assume the *all-in* condition $D^+ \subseteq J$. Then

$$M' = (M - D^-) \,\dot\cup\, D^+ \subseteq \mathcal{F}, \qquad J' = \mathcal{F} - M'$$

is a valid new join, and the residual components and their d_C's stay fixed. For each old residual component define

$$q'_C = q_C - |D^- \cap \delta(C)| + |D^+ \cap \delta(C)|, \quad \Delta_C = q'_C - q_C, \quad \epsilon_C = \Delta_C \pmod 2.$$

Let $\mathcal{O}$ and $\mathcal{E}$ be the old odd and even components, and use subscripts 0 and 1 for the value of ϵ_C. Put

$$S(M,J) = \sum_{C\in\mathcal{O}} k_C.$$

Splitting into the four parity cases gives

$$\begin{aligned} S(M',J') - S(M,J) = &-\sum_{C\in\mathcal{O}_0} \Delta_C - \sum_{C\in\mathcal{O}_1} k_C \\ &+ \sum_{C\in\mathcal{E}_1} (d_C - q'_C). \end{aligned} \tag{6.2}$$

Define the aggregate gain by

$$\mathcal{G}(X) = \sum_{C\in\mathcal{O}_1} k_C + \sum_{C\in\mathcal{O}_0} \Delta_C - \sum_{C\in\mathcal{E}_1} (d_C - q'_C). \tag{6.3}$$

Theorem 6.3 (exact fixed-residual descent gate)**.** *Suppose the old tight-component count is zero and the all-in condition holds. Then, for*

$$\Psi(M,J) = \big(\#\{\textit{tight residual components}\}, S(M,J)\big),$$

the fixed-$\mathcal{F}$ candidate satisfies $\Psi(M',J') <_{\text{lex}} \Psi(M,J)$ if and only if

1. *$\epsilon_C = 0$ for every old even component with $d_C = 4$; and*
2. *$\mathcal{G}(X) > 0$.*

If J realizes the globally optimized value $\Psi^*(M)$, these two conditions imply $\Psi^*(M') < \Psi^*(M)$.

Proof. An old odd component cannot have $d_C = 4$, because the old tight count is zero. With d_C fixed, a new tight component can therefore arise only from an old even four-cut whose parity flips. Conversely, such a flip makes the component odd with $d_C = 4$; the alternative $(k'_C, q'_C) = (1,3)$ is excluded by the same cut-xor argument that gives $k_C \geq 3$, so the new profile is $(3,1)$ and is tight. Hence the new tight count is zero exactly under condition 1. Subject to that condition, both first coordinates vanish, and (6.2) says that the second coordinate decreases exactly when $\mathcal{G}(X) > 0$. Global reoptimization can only improve on the admissible join J'. □

This criterion strictly weakens componentwise monotonicity: losses on a surviving odd component and newly odd components of cut size greater than four are allowed when compensated in the aggregate.

For a partner-free direction, all-in supports form the cycle space of

$$L_t = G - \big(M_t \cup (M_{b+t} - J)\big).$$

Writing $a_C = (M_b \cup M_{b+t}) \cap E(L_t) \cap \delta_G(C)$, the parity part of Theorem 6.3 is feasible for a selected old odd shore Q exactly unless

$$\exists \mathcal{I} \subseteq \{C \in \mathcal{E} : d_C = 4\} : \quad a_Q \bigtriangleup \mathop{\bigtriangleup}_{C\in\mathcal{I}} a_C \in B^1(L_t;\mathbb{F}_2), \tag{6.4}$$

where B^1 is the binary cut space. This is the exact reduced binary coordination obstruction; positivity of $\mathcal{G}$ is a separate integer condition.

The archived 80-vertex control shows that (6.4) is real even under simplicity, girth ten, and cyclic edge connectivity at least four. For one displayed $(5, 1)$ blocker, all three partner-free directions fail: two have $a_Q \in B^1(L_t)$, while the third has $a_Q \triangle a_C \in B^1(L_t)$ for one protected even four-cut. Independent reconstruction checks the graph, flow, cuts, complete fundamental-cycle bases, and every all-in cycle. The same graph has an explicit proper 3-edge-colouring, so it is a Tait control rather than a counterexample in the non-Tait minimal domain. The unconditional result is Theorem 6.2; universal coordination in the non-Tait high-girth case remains open.

7 The direct radius-one assertion

The most direct strengthening suggested by Theorem 5.1 is:

> *For every simple cubic cyclically 4-edge-connected graph and every nowhere-zero W-flow f, either f is H–S-good or one legal simple-cycle switch from f is H–S-good.*

This statement concerns the current H–S component condition, not a fixed five-cover and not an older all-value-class nonpacking condition. The following exact countermodel refutes it.

8 An exact strict 26-vertex countermodel

Graph6 orders the edges lexicographically by second endpoint and then by first endpoint. Consider the graph

`Y??G@cOOGC??P??Ap??¿_A??CIC???IC_C?@??HO??A??B_G????D?_`

and the following flow values in that edge order:

$$\begin{aligned}(5,1,5,4,4,4,7,1,6,7,3,1,6,4,3,2,3,2,6,1,\\ 2,5,5,2,7,6,3,2,5,7,1,1,7,6,5,3,1,7,6).\end{aligned} \tag{8.1}$$

Theorem 8.1 (strict radius-one countermodel)**.** *The graph and flow in (8.1) have the following properties.*

1. *The graph is simple, cubic, cyclically 4-edge-connected, has girth five, and has no Tait 3-edge-colouring.*

2. *The displayed values form a nowhere-zero W-flow with profile*

$$(6, 6, 4, 6, 2, 4, 4).$$

3. *Its value-4 class packs two edge-disjoint boundary joins in its complement.*

4. *The graph has 9,213 simple cycles and exactly 1,485 legal cycle–value switches from this flow. None is H–S-good. The minimum defect at radius at most one is two.*

5. *The graph has the explicit standard FiveCDC given below.*

6. *The displayed flow has exact H–S distance two.*

FiveCDC certificate. Encode a pair of cover coordinates by a five-bit integer. In graph6 edge order, the labels are

$$\begin{gathered}(3,20,5,17,3,5,6,3,3,5,10,6,3,5,3,6,5,9,10,12,\\ 6,18,24,12,20,6,18,6,20,18,6,6,18,20,6,5,5,6,3).\end{gathered} \tag{8.2}$$

Every label in (8.2) has Hamming weight two. For each of the five bit positions and each vertex, an even number of incident labels contain that bit. Thus the five bit-coordinate edge sets are Eulerian and every edge belongs to exactly two of them.

Exact distance-two certificate. Starting from (8.1), perform

$$\begin{aligned} a &= 4 \qquad C = (0,1,2,8,9,20,22,24,25,26,30,31,33),\\ a &= 4 \qquad C = (6,12,14,15,34,35). \end{aligned} \tag{8.3}$$

Both supports are simple cycles and avoid the current value class being added. The successive profiles are

$$(4,6,4,2,4,4,4), \qquad (4,6,4,0,6,4,4). \tag{8.4}$$

Thus (8.3) gives an H–S-good flow after two steps.

Packing certificate. The value-4 matching is

$$M_4 = (3,4,5,13).$$

The two edge sets

$$\begin{aligned} J_0 &= (1,14,15,18,20,21,26,28,29,30,31,32,35,37,38),\\ J_1 &= (0,2,6,7,8,9,12,16,17) \end{aligned}$$

are disjoint, avoid M_4, and both have boundary ∂M_4. The even support furnished by Theorem 5.1 has exactly the two components in (8.3). Hence this certificate explains the distance-two repair while proving that packability alone does not force a radius-one neighbour.

Certificate proof of Theorem 8.1. The supplied standard-library checker decodes graph6 from first principles. It verifies distinct edges, degree three, connectedness, the 26 vertex flow equations, and all 39 nonzero values. It examines every deletion of one, two, or three edges and rejects a deletion as a cyclic cut only after constructing all components and confirming that at least two contain cycles. It independently computes girth and exhaustively backtracks over proper three-edge-colourings.

For completeness of the radius-one enumeration, the checker performs a depth-first enumeration of simple cycles. The minimum vertex of a cycle is its unique allowed starting vertex. The two possible traversal directions are quotiented by retaining the direction whose second vertex is smaller than its final vertex. Hence each undirected simple cycle is generated exactly once. For each generated cycle and each $a \in W - \{0\}$, the checker tests legality directly and, when legal, recomputes the full seven-entry defect profile from graph components. The resulting exact totals are 9,213 cycles, 1,485 legal switches, and zero clean neighbours. This proves $\mathrm{dist}_{\mathrm{HS}}(f) \geq 2$. Direct verification of (8.3)–(8.4) proves $\mathrm{dist}_{\mathrm{HS}}(f) \leq 2$.

Finally, the checker verifies the matching and both join boundaries, and tests every weight and every vertex-coordinate parity in (8.2). These finite checks prove all six assertions. □

Remark 8.2. Theorem 8.1 is a counterexample only to the radius-one auxiliary statement. Its explicit FiveCDC proves that it is not a FiveCDC counterexample.

9 Forty- and sixty-vertex controls

Two controls separate radius-one failure from the existence of a cover and from the packing-to-switch mechanism.

9.1 A Tait-colourable 40-vertex negative control

There is a simple cubic, cyclically 4-edge-connected graph of girth five with a nowhere-zero flow having profile

$$(6, 8, 10, 4, 10, 6, 6). \tag{9.1}$$

The checker exhausts 941,438 simple cycles and 48,544 legal cycle–value switches. None is H–S-good, and the minimum defect at radius at most one is two. In contrast to Theorem 8.1, this graph has a displayed Tait three-edge-colouring; its three colour-pair 2-factors give a three-cycle double cover. This is therefore a clean negative control for the reconfiguration assertion, not for FiveCDC.

9.2 A 60-vertex positive control and the packing lemma

The 60-vertex control is simple cubic and cyclically 4-edge-connected. Its starting profile is

$$(8, 8, 4, 6, 6, 8, 6). \tag{9.2}$$

Switching $a = 7$ on the 25-edge simple cycle

$$\begin{aligned} C = (&1, 2, 8, 11, 18, 22, 24, 26, 27, 30, 31, 33, 35, 37, 39, 40, \\ &42, 43, 45, 62, 65, 67, 73, 74, 75) \end{aligned} \tag{9.3}$$

gives profile

$$(10, 6, 6, 8, 8, 6, 0). \tag{9.4}$$

Thus this state is repaired in one step.

The repair also instantiates Theorem 5.1. For the value-7 matching, two edge-disjoint ∂M_7-joins in $G - M_7$ are

$$\begin{aligned} J_1 = (&1, 2, 13, 16, 18, 20, 26, 32, 35, 44, 46, 49, 50, 53, 57, 60, 61, \\ &62, 63, 64, 69, 71, 72, 73, 75, 81, 83), \\ J_2 = (&3, 6, 7, 10, 12, 14, 17, 19, 22, 24, 27, 29, 34, 37, 40, 41, 43, \\ &45, 47, 48, 55, 56, 74, 80, 82, 85, 86, 87, 88). \end{aligned} \tag{9.5}$$

With $F = F_7(f)$, the checker verifies

$$C = (F - M_7) \triangle J_1, \qquad J_2 \cap (M_7 \cup J_1) = \varnothing. \tag{9.6}$$

Equations (9.3)–(9.6) are therefore a literal connected-support case of the human lemma.

10 A Jaeger-star reciprocal-exchange radius-two obstruction

This section uses a different adjacency relation from the legal simple-cycle switches above. Let r be a cubic vertex with incident edges s_0, s_1, s_2, and partition

$$E(G - r) = A_0 \sqcup A_1 \sqcup A_2$$

Table 1: The three principal exact computations. "Legal" counts cycle–value pairs, not merely cycles.

order	cycles	legal	initial minimum	radius-one minimum	outcome
26	9,213	1,485	2	2	packs; no one-switch repair
40	941,438	48,544	4	2	no repair; Tait control
60	—	—	4	0	displayed one-switch repair

so that

$$T_i = \{s_i\} \cup (E(G - r) - A_i)$$

is a spanning tree for each i. Let $K_i \subseteq T_i$ be the unique edge set having odd degree at every vertex, and define

$$f(e)_i = 1 \quad \iff \quad e \notin K_i.$$

A *reciprocal exchange* swaps one edge of A_i with one edge of A_j, provided both changed sets still define trees. We order states lexicographically by

$$\Psi = \left(\min_{\mu \neq 0} d_\mu(f), |K_0| + |K_1| + |K_2| \right).$$

The auxiliary search also declares a state terminal when one of 21 parallel whole-circuit span systems certifies a switch to an H–S-good flow. This is more permissive than direct H–S goodness and is kept separate from it below.

Proposition 10.1 (exact reciprocal-exchange distance three)**.** *There is a simple cubic, 3-edge-connected, cyclically 4-edge-connected, nonplanar graph of order 40 and girth five with a rooted Jaeger-star state such that:*

1. *its kernel sizes are* $(20, 20, 20)$*, its H–S defect profile is* $(4, 6, 2, 6, 6, 6, 2)$*,* $\Psi = (2, 60)$*, and all 21 parallel span flags fail;*
2. *no lower-*Ψ*, H–S-good, or parallel-span-successful state is reachable by at most two legal reciprocal exchanges; and*
3. *a displayed sequence of three reciprocal exchanges reaches an H–S-good flow.*

The same graph has an explicit standard FiveCDC.

Exact data. The labeled graph6 record used for the state is

```
ghe?GC@@G??@?@??_@I??A?C_?G??G@?CA?@?C?G_??_@?@???@?@??_?C?G??C@?O??C?A??G????G????C????@??O??G?????_????H???
?_@?????G_????@G??_?C@.
```

Nauty/Traces `labelg -q` gives the canonical record

```
gs?GO???G?_????D?@_?G?ACC?OC@?GC?A@G?O??A???CO??G?O???c??O_????O???@C???A_???Q????__@????A?????_???Ga????WG???
?GG????CG????CI????OC.
```

At root 0, in labeled graph6 edge order after deleting its three incident edges, the omitted-class masks are

$$(74750917167123596,\ 60249003513889361,\ 91152673948429\,14).$$

For each $\mu \neq 0$, the checker evaluates all four matching values r with $\mu(r) = 1$, exactly as in Theorem 3.16 of [3]. At the seed the resulting component-defect matrix is

$$\begin{pmatrix} 4 & 4 & 4 & 4 \\ 6 & 6 & 6 & 6 \\ 2 & 2 & 2 & 2 \\ 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 \\ 2 & 2 & 2 & 2 \end{pmatrix}.$$

Thus no zero profile entry is hidden by a choice of affine matching value.

Exhaustive radius-two certificate. There are $3 \cdot 19^2 = 1083$ candidate first swaps and exactly 93 legal neighbors. None is lower or successful; 24 are safe with the same Ψ. Exhausting every legal neighborhood of those 24 states checks 2196 second arcs, with no lower Ψ, no zero H–S defect, and no successful parallel span flag. These arcs have 1895 distinct targets, 1852 of them new at distance two.

The local-position swaps

$$(10, 9), \qquad (40, 43), \qquad (15, 17)$$

exchange full graph edges

$$(13, 12), \qquad (43, 46), \qquad (18, 20).$$

The successive exact data are

state	kernel sizes	profile	Ψ	span flags
seed	$(20, 20, 20)$	$(4, 6, 2, 6, 6, 6, 2)$	$(2, 60)$	0
1	$(20, 20, 20)$	$(4, 6, 2, 6, 6, 6, 2)$	$(2, 60)$	0
2	$(20, 20, 20)$	$(4, 6, 4, 6, 2, 4, 2)$	$(2, 60)$	0
3	$(20, 20, 21)$	$(6, 6, 6, 6, 4, 4, 0)$	$(0, 61)$	3

The zero in the last profile is direct Theorem 3.16 component-parity success for the current flow. The three span flags are a separate additional fact. Hence the exact augmented reciprocal-exchange distance is three.

FiveCDC-positive caveat. In labeled graph6 edge order, the following five-bit labels all have weight two:

$$\begin{aligned} &(20, 24, 9, 5, 3, 17, 24, 18, 10, 9, 3, 5, 20, 5, 6, 12, 12, 6, 20, 17, \\ &\ 18, 3, 9, 10, 10, 3, 10, 9, 10, 3, 3, 9, 10, 17, 3, 10, 18, 9, 24, 24, \\ &\ 17, 17, 9, 24, 10, 18, 9, 17, 24, 24, 20, 9, 12, 3, 5, 12, 6, 24, 18, 10). \end{aligned} \tag{10.1}$$

Their xor is zero around every vertex. Therefore their five bit coordinates are Eulerian and cover every edge exactly twice. This proves directly that the graph is FiveCDC-positive.

Proof. The independent standard-library checker decodes both graph and state, checks the tree and odd-kernel definitions, exhausts all zero-, one-, two-, and three-edge cuts, and computes girth. It then performs the literal finite enumerations and parity calculations just stated. Nauty/Traces `planarg -v` independently reports the graph as nonplanar. Direct vertex-coordinate parity checks verify the displayed FiveCDC. The exhaustive lower bound through radius two and the literal three-exchange path prove the exact distance. □

Remark 10.2. Proposition 10.1 concerns reciprocal exchanges of three-tree states. It is not a statement about the simple-cycle-switch distance $\mathrm{dist}_{\mathrm{HS}}$ used in Theorem 8.1, and it does not imply that this graph is hard for every flow. Its role is solely to invalidate a project-created radius-two shortcut.

11 Verification architecture

The principal checker is

`scratch/verify_husek_samal_packable_one_switch_countermodel.py`.

It uses only the Python standard library. Its graph6 decoder, flow checker, component-defect routine, simple-cycle generator, switch routine, cyclic-cut enumeration, girth computation, Tait-colouring backtracker, and FiveCDC checker are implemented locally. No SAT solver or opaque mathematical library is used.

The 40-vertex negative control is checked by the separately written

`scratch/verify_husek_samal_one_switch_boundary.py`.

The 60-vertex construction has a second checker,

`scratch/verify_fano_order60_flow_repair.py`.

It reconstructs the graph from two 32-vertex pole recipes, verifies the one-switch repair and the two joins, and separately follows a 13-switch path to the flow induced by an explicit FiveCDC. The two programs share no imported project module.

The reciprocal-exchange obstruction is frozen in

`scratch/jaeger-order40-augmented-radius-two-counterexample-20260729/`.

Its independently written Python checker reconstructs every first and second neighborhood, all 28 Theorem 3.16 defects, the three-step path, and the FiveCDC certificate. Its verifier SHA-256 is

`7c4a9d165416392399aa53ca6226573f459bae07e31dd1a76332b636d0b1be17`.

The rank-three distance-five pair is checked twice. The source-side standard-library verifier reconstructs the graph, both flows, all 168 three-move orders, all 1848 four-move orders, the five-move path, its 63 circuit components, and all 2640 legal reciprocal-exchange neighbours. A separately written checker rederives Theorem 4.1, tests its component criterion against 16384 loop/parallel-edge instances, and repeats the 1848-order classification using two different affine lifts.

The frozen artifact hashes at the time of this draft are:

SHA-256	file
`9bdcb3a65f6dc07224f8a20c79e45e9085c9f73c55ead745ac64127decd76607`	packable-state checker
`ac51aaf266178eb764fcada85b25e17dc2c18c129413abb51a2b309513ea415d`	boundary checker
`09f9bdc797948b49cf6b7a3080160e07111dad718f0e07ae00a2a9c0958b9a6e`	independent 60-vertex replay
`5cd280e730e0dc336681c3768945b7c691e023cb4f957a7408ea306597053f45`	packable order-26 retained state
`6e1dc5fb3b290fe8364559844ed419ee6dda1ec974b2ad3aebfd21442c8fe6aa`	order-40 retained state
`450a066619b5b21f7bf64ca239b9f2798219af6b695ee619bfbf844883e25476`	distance-five certificate
`2cc3ca1397b08ed8ba555c569d2aaf13e57354a5e847d6447cd6242d1c4530a2`	distance-five source checker
`c0021e2c4cc081918a9edcdb8f1925f9d9c858cf48861d9577178ac952d9a9cf`	independent rank-three checker

From the project root, the exact replay commands are

```
python3 scratch/verify_husek_samal_packable_one_switch_countermodel.py
python3 scratch/verify_husek_samal_one_switch_boundary.py
python3 scratch/verify_fano_order60_flow_repair.py
(cd scratch/bpr-second-coordinate-descent-20260731 && ./run_all.sh)
(cd scratch/bpr-second-coordinate-descent-blind-audit-20260731 && \
  python3 -B independent_checker.py)
(cd scratch/bpr-second-coordinate-refined-gate-20260731 && ./run_all.sh)
python3 \
  scratch/jaeger-order40-augmented-radius-two-counterexample-20260729/verify.py
python3 \
  scratch/petersen-foster-girth10-nontait-20260729/verify_reconfiguration.py
python3 scratch/rank3-fourmove-normal-form-20260729/verify.py
```

12 What remains open

The finite countermodels show only that radius one is not a universal domination bound for H–S-good flows, even in a strict reduced-looking class. They leave open all of the following:

1. whether the reduced cut-space equations and positive aggregate gain can always be coordinated in the non-Tait, cyclically-4, girth-ten minimum-counterexample domain;
2. whether every nowhere-zero $\mathbb{F}_2^3$-flow can reach an H–S-good flow by some finite legal cycle-switch sequence;
3. whether a larger universal radius exists;
4. whether every bridgeless graph has some initially H–S-good flow, the equivalent formulation of FiveCDC in [3]; and
5. the standard FiveCDC conjecture itself.

The 26-vertex state has exact distance two, but this is one instance, not evidence of a universal radius-two theorem. It also shows sharply that the finite sequence in the packing-to-switch lemma cannot always be replaced by one circuit. The lemma supplies no universal packing theorem.

Proposition 4.2 rules out a four-move bound only for the broader fixed-value Eulerian-support distance between one selected pair. It neither computes the distance from its starting flow to the set of all H–S-good flows nor gives exact connected-cycle distance. In particular, it is not a counterexample to eventual cycle-switch reachability.

Likewise, Proposition 10.1 disproves a universal radius-two assertion only in the separate reciprocal-exchange state graph. It gives no universal radius-three bound and no reduction of FiveCDC to such a bound.

13 Novelty and responsibility boundary

The cited papers establish the H–S component criterion and the general cycle-supported reconfiguration framework. In particular, Esperet et al. prove the general flow decomposition into cycle-supported flows and a complementary-projection connectivity lemma [2]; Cranston et al. also introduce an auxiliary adjacency permitting even support [1]. Over $\mathbb{F}_2^3$, our move has the same fixed-value pattern as standard cycle adjacency but permits a disconnected Eulerian support. Relative to the auxiliary even-support adjacency of Cranston et al., it is narrower in value pattern. We make no novelty claim for either. The elementary packing-to-switch bridge, the exact radius-one formulation, the partner-free shore theorem, the exact aggregate-gain gate, the rank-two legal-order criterion, the rank-three four-move criterion and certified distance-five pair, the reciprocal-exchange obstruction, and the finite certificates may be new, but that assessment is provisional. The radius-two reciprocal assertion was formulated inside this AI-assisted project rather than in the cited literature. The search and literature review were AI-assisted and not exhaustive in the scholarly sense. Priority and publishability require review by specialists in cycle double covers, nowhere-zero flows, and flow reconfiguration.

The human part of Theorem 5.1 is fully displayed. The finite theorems reduce to explicit data and short, inspectable exhaustive programs, but those programs and this prose were generated with AI assistance. Reproducibility is not peer review, and implementation independence is not human independence.

References

[1] Daniel W. Cranston, Jiaao Li, Bo Su, Zhouningxin Wang, and Ningyan Xu. Reconfiguration of nowhere-zero flows, 2026.

[2] Louis Esperet, Kevin Hendrey, Aurélie Lagoutte, Margaux Marseloo, Sergey Norin, and Raphael Steiner. Nowhere-zero flow reconfiguration, 2025. Version 4, revised July 2026.

[3] Radek Hušek and Robert Šámal. Exponentially many circuit double covers, 2026.